

Visualisations

Predicting Financial Wellbeing Among Kenyan Adults

DataSprint 2026 • Multiclass classification of financial status

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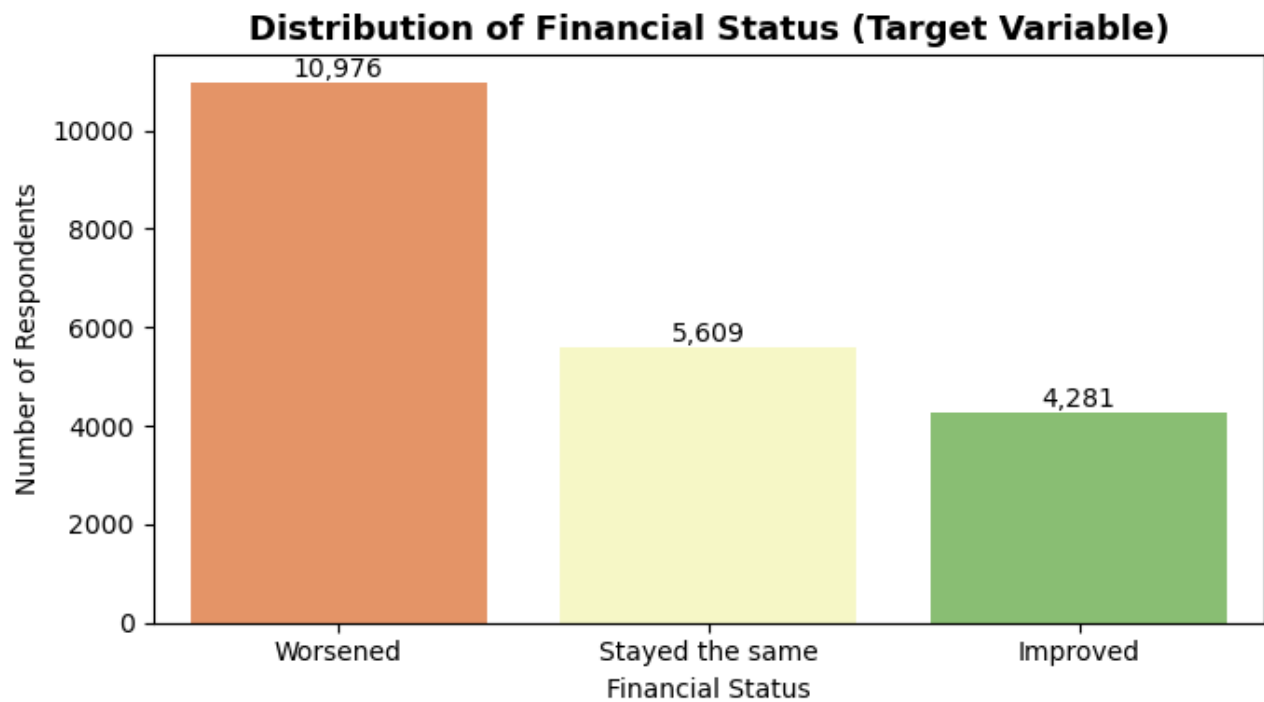
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Part A — Analysis & EDA Visualisations

Exploratory data analysis and model diagnostics generated in the analysis notebook.

FIGURE 01

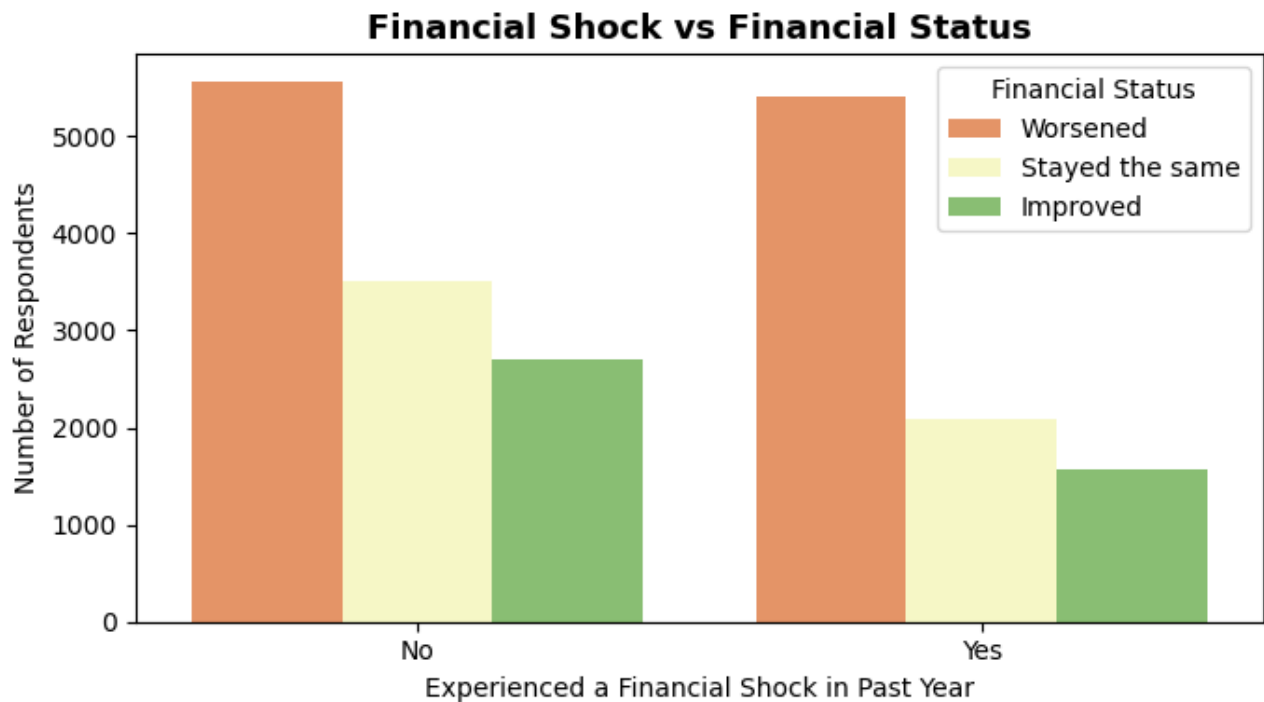
Distribution of Financial Status (Target Variable)



The three-class target. Over half of respondents report their financial situation worsened, making this an imbalanced classification problem.

FIGURE 02

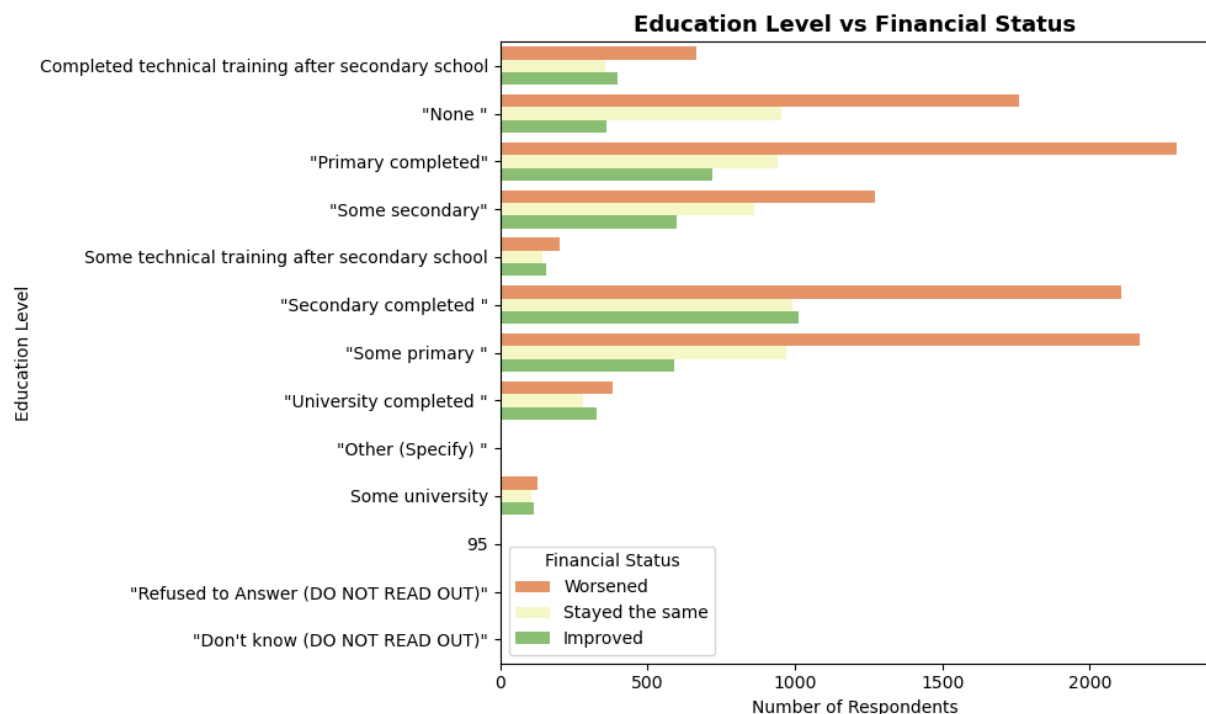
Financial Shock vs Financial Status



Experiencing a financial shock is one of the strongest predictors of worsening — shock respondents skew heavily toward the 'Worsened' class.

FIGURE 03

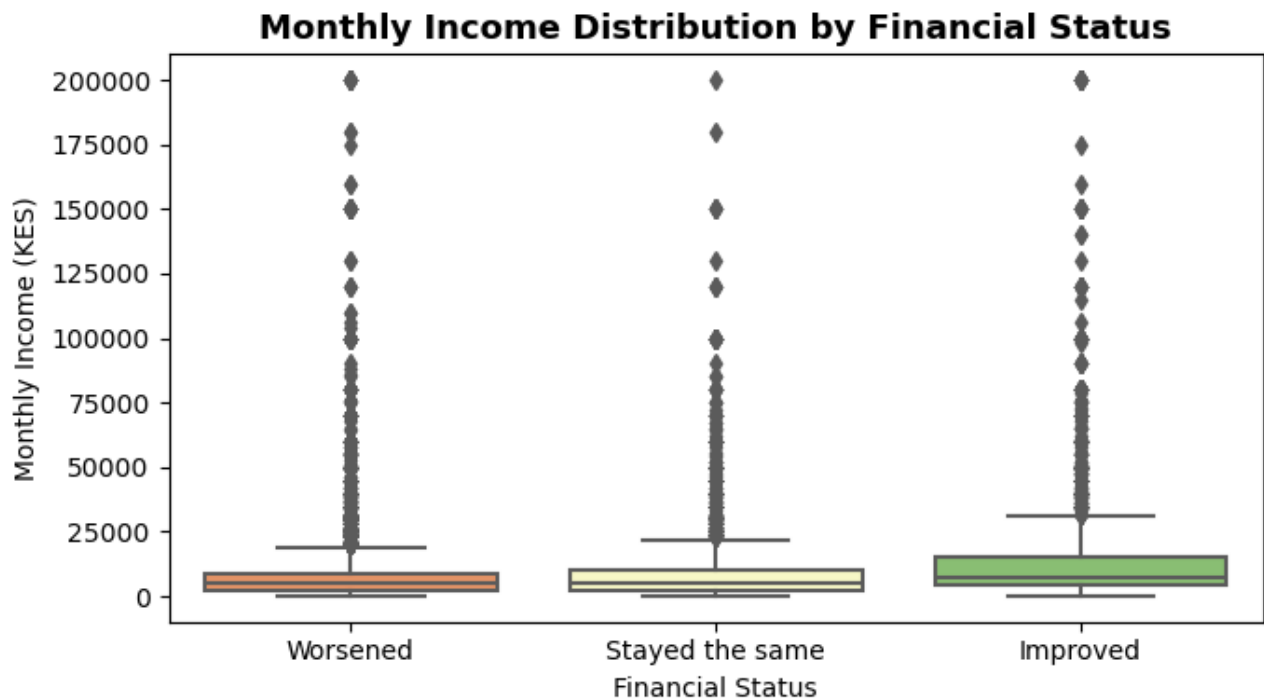
Education Level vs Financial Status



Higher education correlates with better outcomes, but no level is immune. Behaviour and shocks matter beyond education alone.

FIGURE 04

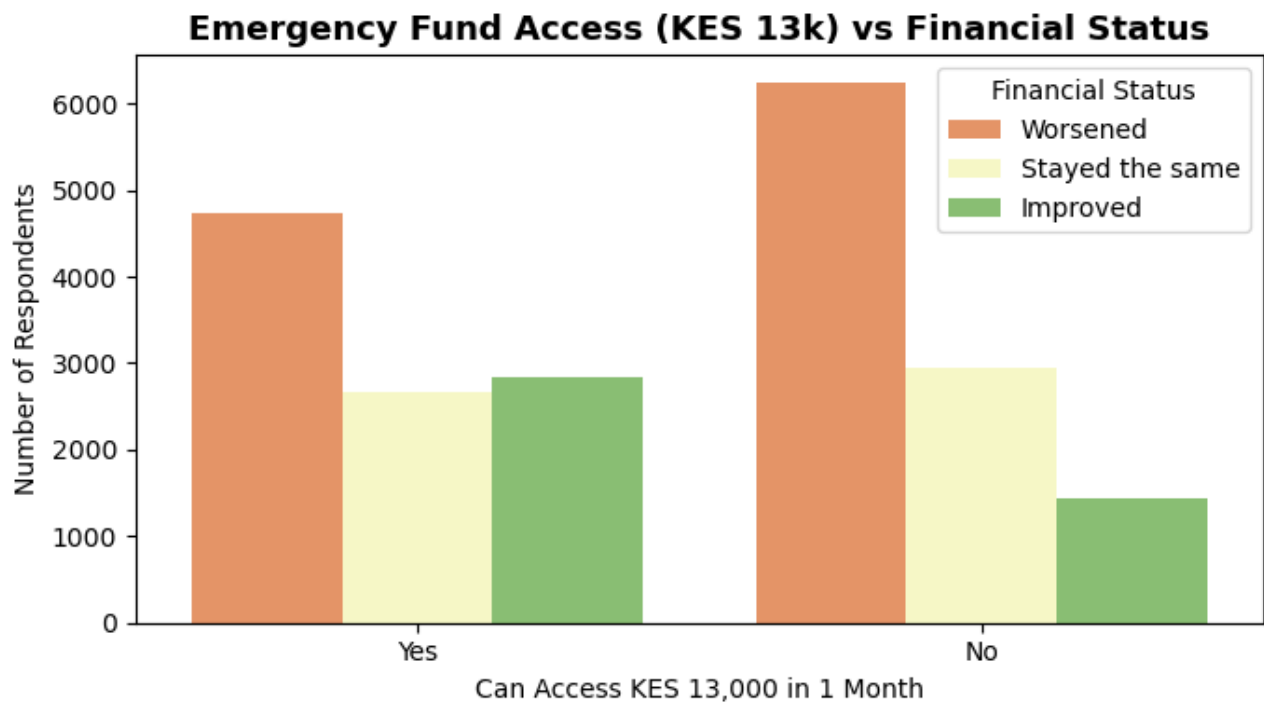
Monthly Income Distribution by Financial Status



'Improved' respondents have a higher median income, but the wide overlap shows income is not destiny — resilience matters more.

FIGURE 05

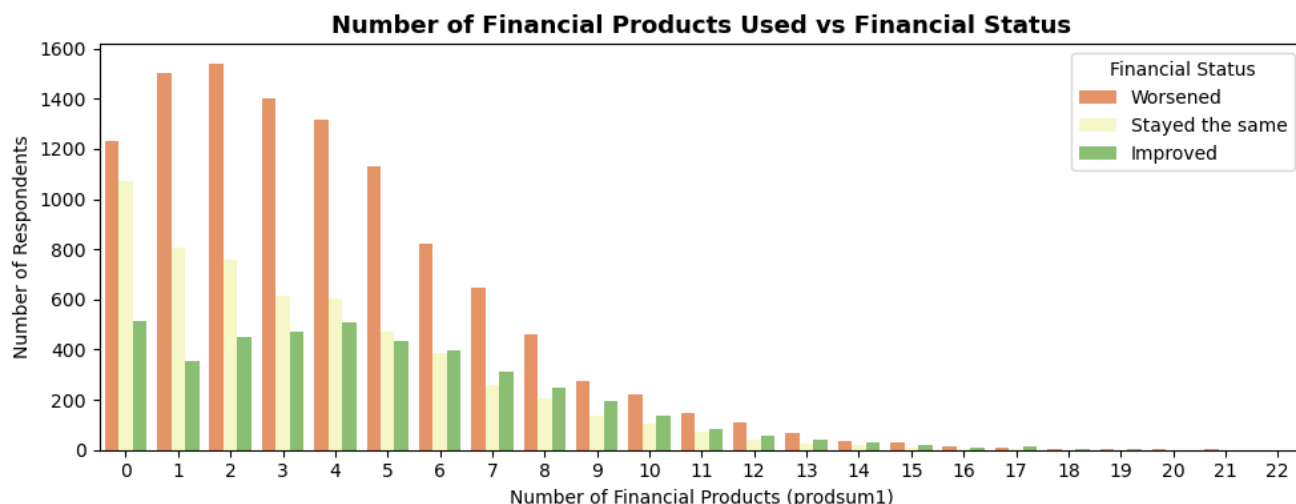
Emergency Fund Access (KES 13k) vs Financial Status



The sharpest divide in the dataset. Those who cannot access emergency funds concentrate almost entirely in 'Worsened'.

FIGURE 06

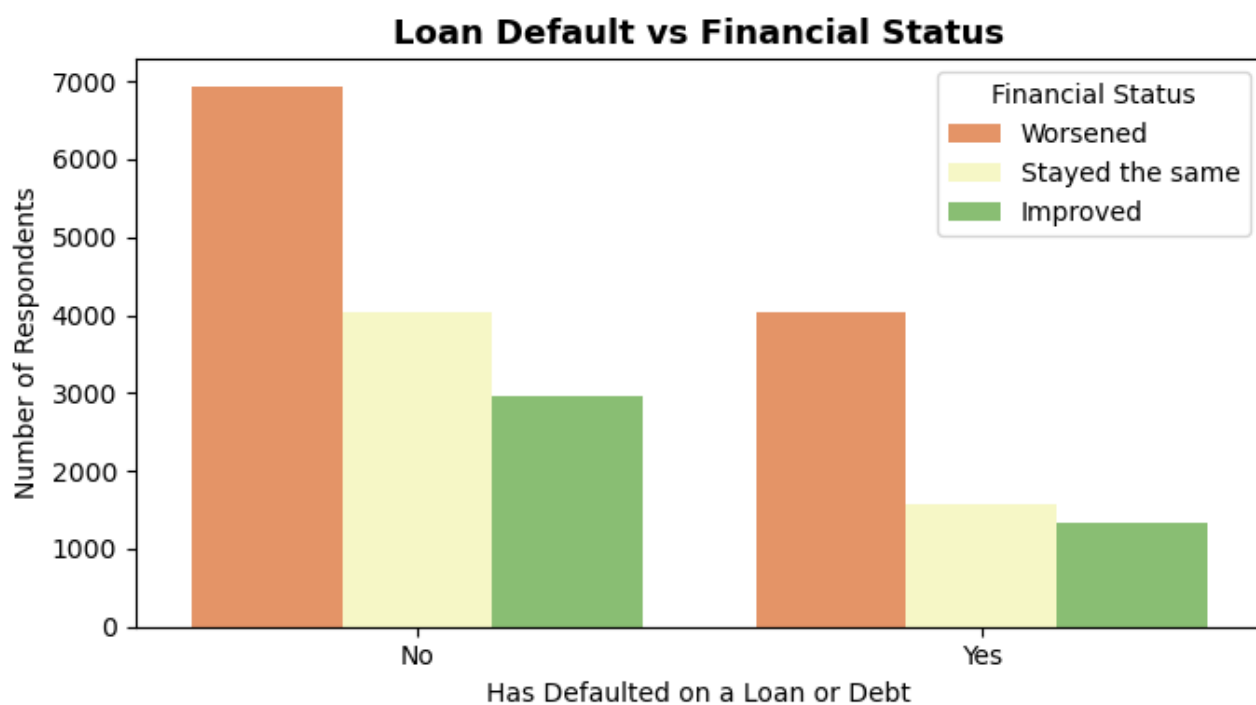
Number of Financial Products Used vs Financial Status



A clear gradient — more financial products used corresponds to better outcomes. Active inclusion builds resilience.

FIGURE 07

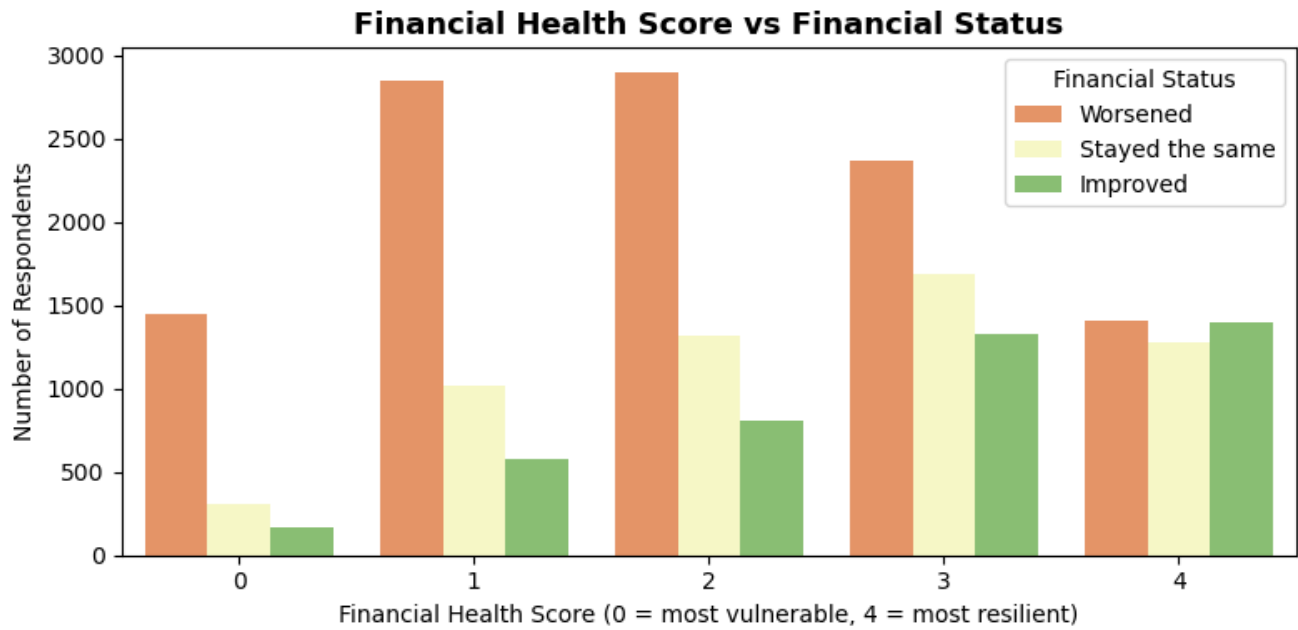
Loan Default vs Financial Status



Default is both a cause and a symptom of financial deterioration; defaulters are heavily concentrated in 'Worsened'.

FIGURE 08

Financial Health Score vs Financial Status (Engineered Feature)



Validation of the engineered resilience score (0 = most vulnerable, 4 = most resilient) against the target.

FIGURE 09

Logistic Regression — Confusion Matrix



Baseline model. Confusion matrix on the held-out test set across the three outcome classes.

FIGURE 10

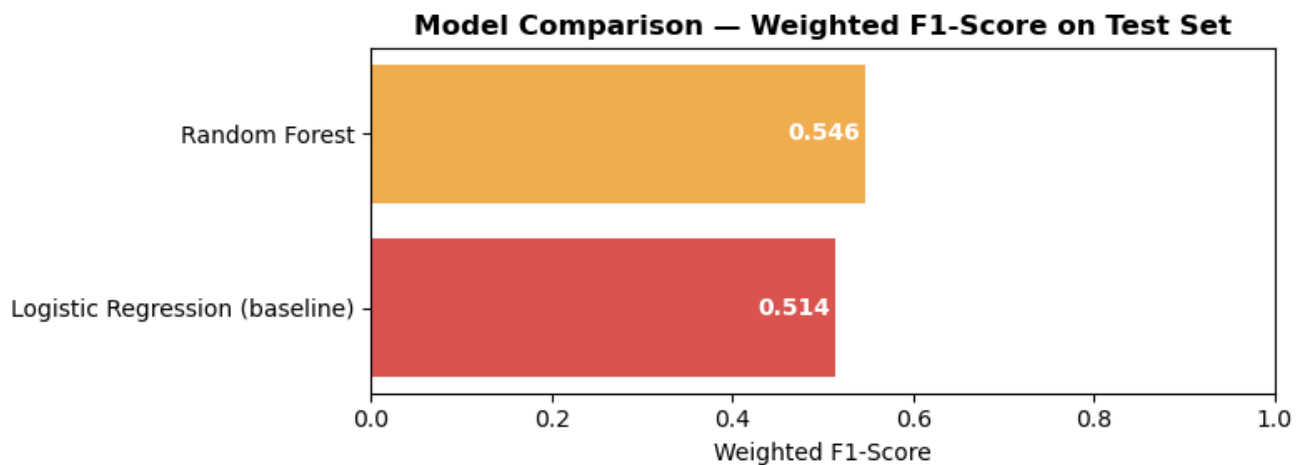
Random Forest — Confusion Matrix



Tuned Random Forest. Stronger diagonal, particularly for the majority 'Worsened' class.

FIGURE 11

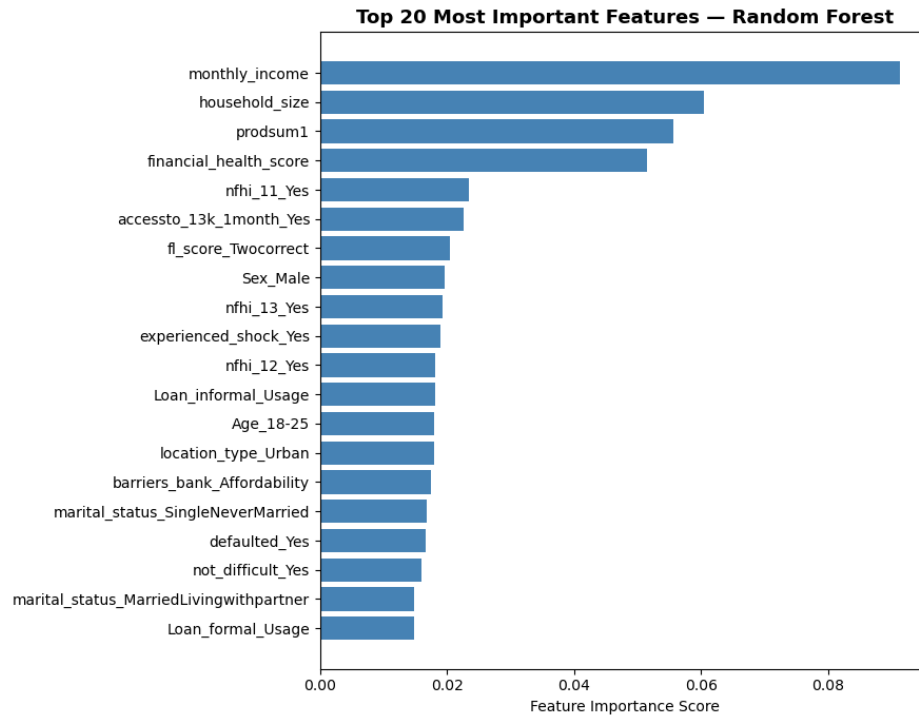
Model Comparison — Weighted F1-Score



Random Forest outperforms the Logistic Regression baseline on weighted F1 over the test set.

FIGURE 12

Top 20 Most Important Features — Random Forest



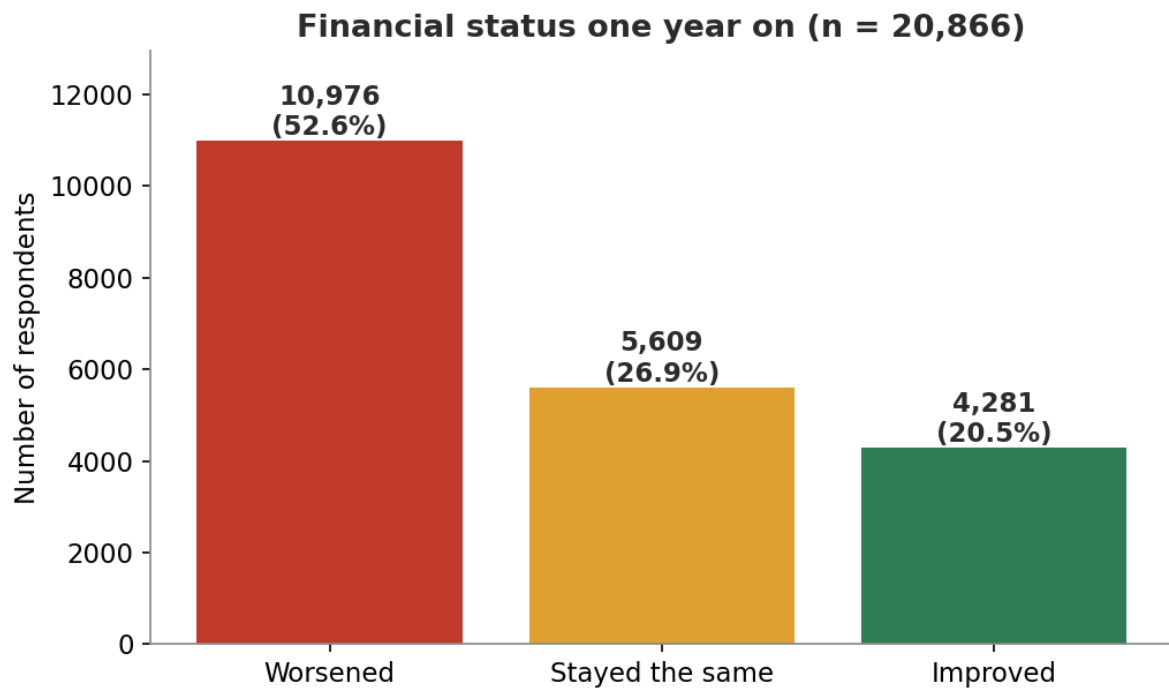
Drivers of the prediction. Monthly income, household size, product usage and the engineered health score lead.

Part B — Presentation Visualisations

Polished, audience-ready charts used in the DataSprint presentation deck.

FIGURE 13

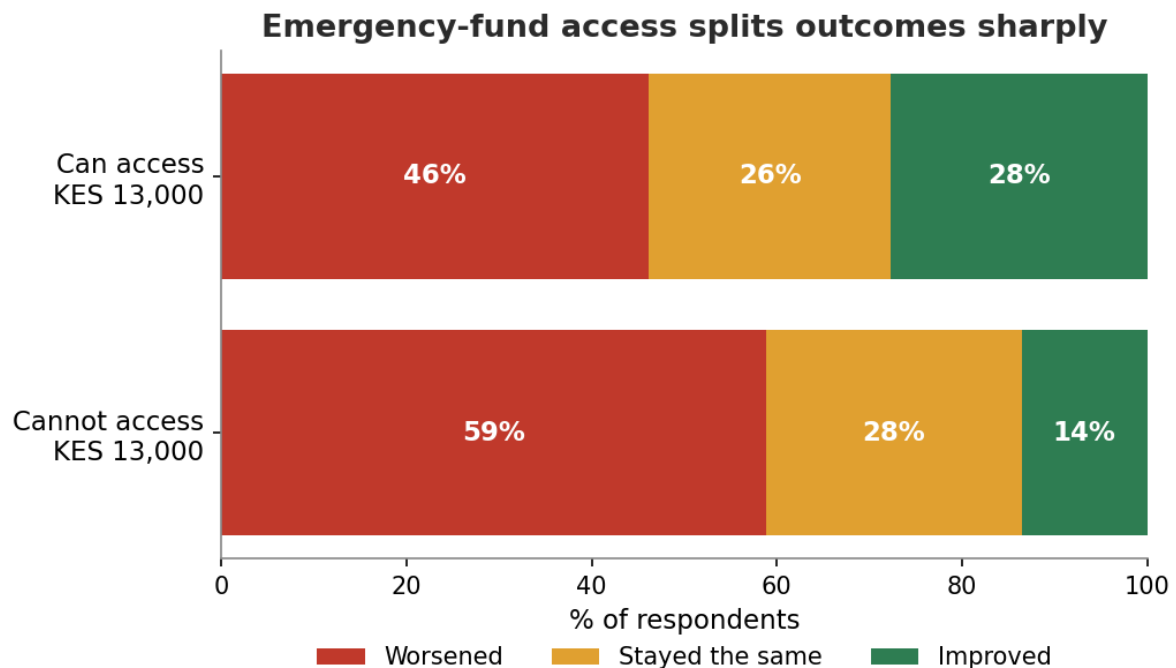
Financial Status One Year On (n = 20,866)



Presentation view of the target distribution with class percentages.

FIGURE 14

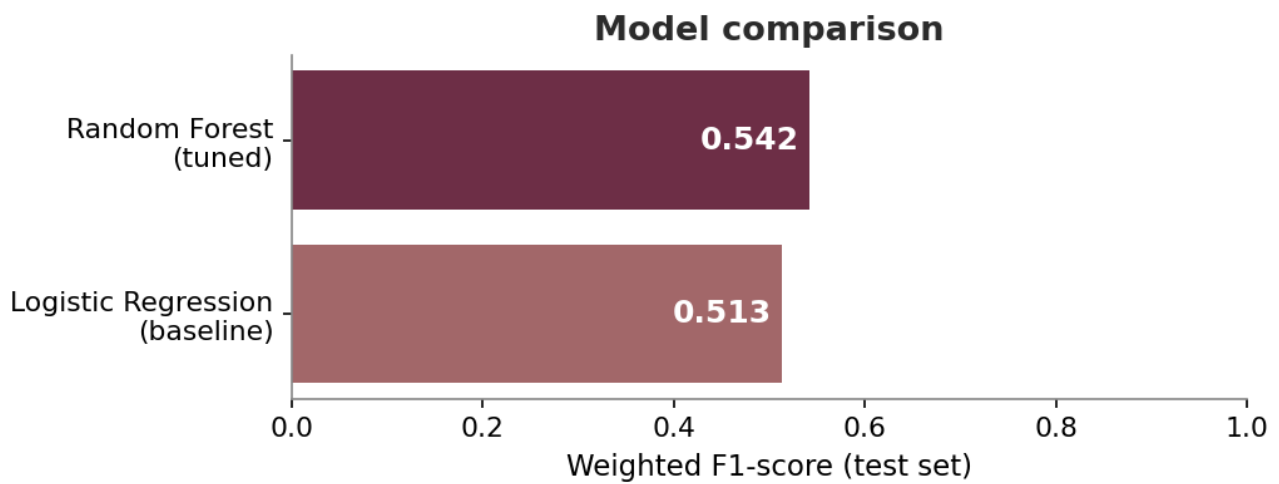
Emergency-Fund Access Splits Outcomes Sharply



100% stacked view: respondents who cannot access KES 13,000 are far more likely to have worsened.

FIGURE 15

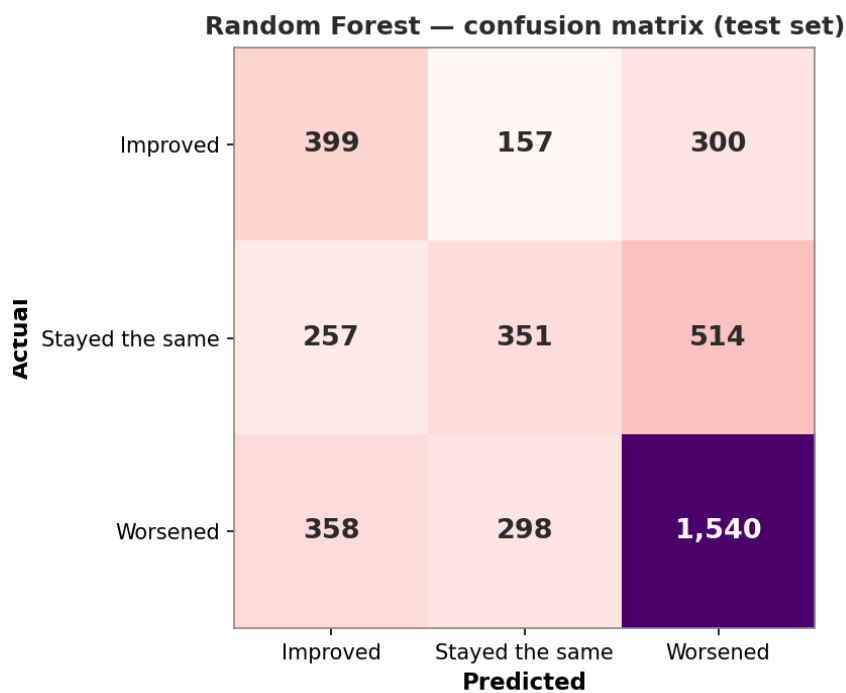
Model Comparison — Weighted F1 (Test Set)



Final scoreboard: tuned Random Forest (0.542) vs Logistic Regression baseline (0.513).

FIGURE 16

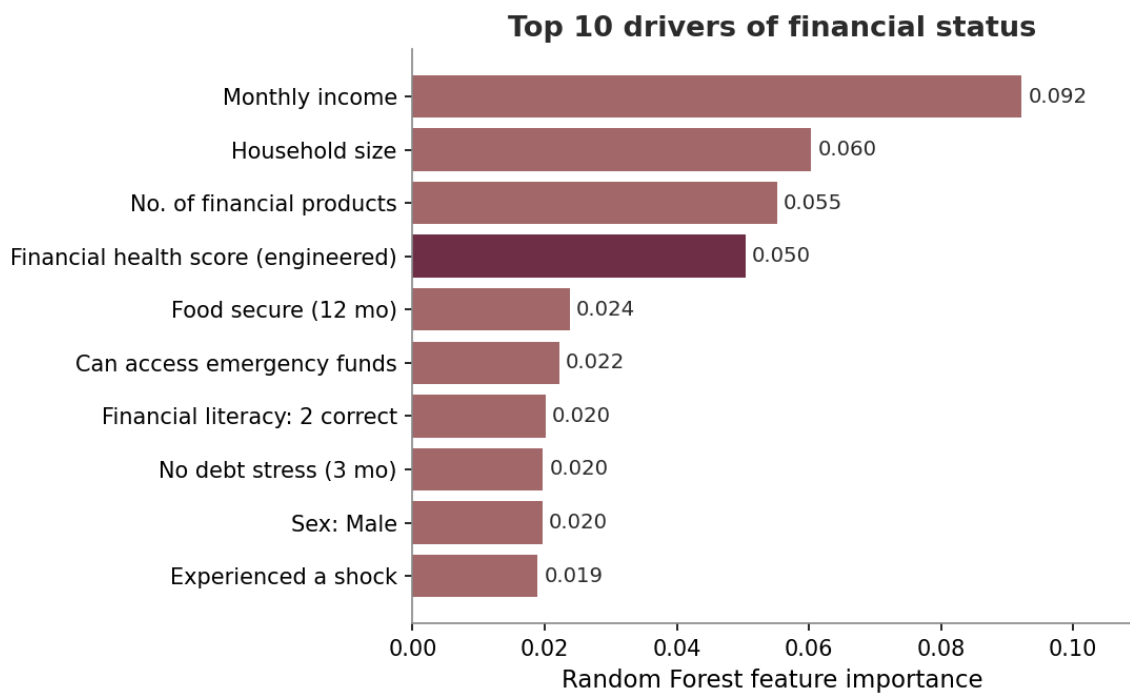
Random Forest — Confusion Matrix (Test Set)



Presentation-styled confusion matrix for the selected model.

FIGURE 17

Top 10 Drivers of Financial Status



Headline feature-importance chart used in the deck, with human-readable feature labels.